Research Method of Data Science

Week 9: Finding and Validating Quantitative Data

Iegor Vyshnevskyi, Ph.D.

Sogang University

October 27/29, 2026

Week 9: Finding and validating quantitative data

Tuesday: how to judge and document a data source

Thursday: import, inspect, document, and validate

Companion notebook: **W9_RMDS_Data_Source_Lab.ipynb**

This week's output: a documented source decision for your individual research project.

Agenda

Tuesday — Week 9, Meeting 1

1. Finding a source that fits your question
2. Provenance, metadata, and revisions
3. Licensing and access through downloads and APIs
4. Validation and source decisions
5. Preparation for Thursday

Appendix: source log, checklist, software, and references

After the midterm: the next research decision

Week 7: **What evidence would answer your question?**

Week 9: **Does the available dataset actually supply it?**

Think, pair, share — 3 minutes

Name one variable your project needs. What would you check before trusting a column with that name?

1. Finding a source that fits your question

A search specification narrows the task

Before searching, write:

- Question and intended claim
- Unit, population or geography, and observation period
- Concept and acceptable measure or proxy
- Required frequency and comparisons

Example: **district-level household waste and recycling, annually, 2021–2024**.

A proxy needs a reason and a stated limitation.

Discovery and evidence have different roles

Starting point	What to do next
Literature article	Trace its dataset, sample, definitions, and access
Data portal	Find the original table and producer
Dataset search / repository	Check provenance, version, documentation and rights
AI search suggestion	Locate and verify the actual source

A search result or AI answer cannot establish a dataset's coverage.

> provenance is the documented origin and history of the evidence, not a guarantee of accuracy.

Useful places to begin

Research need	Examples of starting points
Korea population, labour, regions	KOSIS
Cross-country development indicators	World Bank data
Economic and financial series	FRED
Historical economic data	OECD Data
International trade data	UN Comtrade
A published study's data	Journal archive or linked repository

Check the specific table's definitions and rights. Access conditions vary.

Fit means more than a familiar column name

Check	A possible mismatch
Concept / measure	Foreign direct investment inflows versus Multinational Corporation manufacturing activity
Unit / geography	National totals versus provincial comparisons
Population / coverage	Platform listings versus all housing transactions
Time / frequency	Annual averages versus a policy change within a month
Definitions / units	Income versus consumption; current versus constant prices

Record the mismatch before choosing an analysis.

Source choice challenge — pairs, 4 minutes

Question: How did household waste per registered household vary across districts, 2021–2024?

A: annual district waste totals and registered households, with documentation.

B: national recycling percentage, 2010–2025.

C: a survey of volunteers' environmental attitudes this month.

Choose a candidate. Identify one further check before accepting it.

> Hint: the question asks for a district-level measure, not a national average or a survey of attitudes.

2. Provenance, metadata, and revisions

Provenance records where the data came from

Provenance: the documented origin and history of the evidence.

For a downloaded series, distinguish:

- **Producer:** who collected or estimated it
- **Distributor:** where you obtained it
- **Your actions:** filters, downloads, conversions and later changes

Example: original statistical agency, portal distribution, your country-year selection.

Metadata make the numbers interpretable

Metadata: information describing the dataset and its fields.

Field	Why it matters
Unit and definition	What exactly does the number represent?
Geography and population	Who or what is covered?
Frequency and period	Are observations comparable in time?
Missing and suppression codes	Does a symbol mean unknown, withheld, or zero?
Method and revision notes	Has construction or coverage changed?

Four dates that can differ

Observation period: when the measured activity occurred.

Release date: when the producer published the estimate.

Update/version date: when the distributed dataset changed.

Retrieval date: when you obtained your copy.

A download today can contain revised estimates for earlier years.

A source log and a dictionary do different jobs

Source log

Provider, title, series/table ID, URL, version, retrieval date, filters, access terms, file location.

Data dictionary

Variable, definition, unit, type, allowed values, missing codes, coverage notes.

Keep both beside the notebook.

Metadata puzzle — pairs, 4 minutes

A source contains:

```
district_code = NA
year = 2024
recycling_pct = 0.42
```

1. Could NA be an identifier?

2. Does 0.42 mean 0.42% or 42%?

3. Which documentation would resolve these questions?

> Hint: the codebook specifies fraction vs percentage. The core dataset uses valid NA code and 0–100 scale. A plausible range is not sufficient to identify correct scale.

3. Licensing and access through downloads and APIs

Access and sharing are separate decisions

For each source, record:

- License or usage terms and the original link
- Attribution required by the provider
- Restrictions on reuse or redistribution
- Access conditions for confidential or licensed records

Unknown terms: record the uncertainty and clarify before redistribution.

An example of source specific terms

The World Bank's GDP per capita indicator page displays:

GDP per capita (constant 2015 US$)

Indicator: `NY.GDP.PCAP.KD`

License shown: CC BY 4.0 It means you are free to share and adapt the data, as long as you give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the World Bank endorses you or your use.

The dataset terms describe attribution and exceptions for some third-party data.

Save the terms relevant to the actual series you use.

The access route should preserve a clear record

Route	What you retain
CSV / Excel download	Original file, filters, table/series ID, date
API request	Exact query, all response pages, metadata, date
Permitted HTML extraction	Page URL/date, extraction rule, coverage checks
Approved restricted access	Permission and reconstruction/access explanation

Week 7 introduced the routes. This week checks their outputs.

An API response needs inspection

API: a documented interface for requesting structured data.

A request specifies countries, an indicator, dates, and a format.

A response may include:

- metadata and records
- **pagination**: records split across several response pages
- null values or an error message

HTTP success does not guarantee a complete research dataset.

API audit — pairs, 5 minutes

A request covers **2 countries × 5 years** for one indicator.

The response says:

```
page: 1    pages: 2    total: 10
records in this response: 5
```

What must happen before you claim complete coverage?

What if the final 10 records contain two missing values?

> Hint: check the unique country-year-indicator keys against the expected grid, rather than trusting row count alone.

4. Validation and source decisions

Validation compares data with explicit expectations

A **validation rule** states what should hold and why.

Examples:

- Expected columns and types follow the dictionary.
- A district-year key should identify one record.
- Recycling percentage uses a 0–100 scale.
- Every expected district-year appears.

Some failures require source clarification rather than an immediate correction.

A validation checklist

Layer	Example check
Structure	Required columns and numeric parsing
Identity	Missing or repeated observation keys
Values	Units, valid categories, permissible ranges
Coverage	Missing groups, years, or observations
Consistency	Compatible definitions, totals, and frequencies
External comparison	Match a documented value from the same source/version

> The checklist is a guide, not a guarantee of correctness.

The notebook dataset has deliberate problems

Fictional district environment records, 2021–2024

One row should represent one district-year.

- Three districts, four expected years
- Household counts, waste tonnes, and recycling percentage
- `NA` is a valid district code
- `..` means suppressed; `-999` means unavailable waste

The notebook includes the table and full source note.

Validation can pass while the source still fails

A table can have valid types and ranges but still:

- measure an unsuitable proxy
- exclude important groups
- change definitions across years
- reflect revised rather than historically available values

Technical checks support source evaluation. They do not replace it.

A source decision follows the evidence

Accept: fit and checks support the planned use.

Accept with conditions: specify remaining issues, restrictions, and required action.

Reject for this question: explain the mismatch and identify an alternative.

An issue log records **evidence, research consequence, next action, and status**.

5. Preparation for Thursday

What to bring

1. Your individual research question and Week 7 design note
2. One candidate source with documentation and usage terms
3. A small, non-sensitive CSV export, if available
4. A personal copy of **W9_RMDS_Data_Source_Lab.ipynb**

If access is blocked, bring the metadata and a clear explanation.

Everyone can complete the practice workflow without a separate data download.

Thursday's deliverable

Your completed notebook and a **project source audit**:

- Source log and data dictionary
- At least three applicable validation checks with interpretation
- Issue log and a reasoned readiness decision
- One revision after peer feedback

If data are unavailable, document the access obstacle and next action using the practice table for code.

Thursday – Data source lab

Agenda

Thursday — Week 9, Meeting 2

1. Import and inspect the practice data
2. Document and validate the records
3. Apply the workflow to your project
4. Peer check and revision
5. Save and submit

The optional API section is for later exploration.

Setup — 5 minutes

Open **W9_RMDS_Data_Source_Lab.ipynb** in Colab or Jupyter.

In Colab: **File → Upload notebook**, then save a personal copy.

Run Section 1. You should see Python/pandas versions and the working folder.

The core lab uses embedded fictional data and does not require internet requests.

Import and inspect — 10 minutes

Notebook Sections 2–3

1. Read the source note before running the import.
2. Compare default parsing with the preserved import.
3. Inspect shape, identifiers, sample records, and types.
4. Answer Checkpoint 1 in your own words.

Checkpoint: 12 rows, 5 columns, and a preserved `NA` identifier.

Document the fields — 8 minutes

Notebook Section 4

Inspect the data dictionary.

Explain to your neighbour:

- What does one row represent?
- What are the units and valid ranges?
- What do the missing markers mean?
- What can this table tell us about individual households?

Validation lab — 15 minutes

Notebook Sections 5–7

1. Run the typed working-copy and validation cells.

2. Inspect records behind the failed checks.

3. Compare row count with unique-key coverage.

4. Complete Checkpoint 2.

A failed data rule should appear in the report without stopping the notebook.

An issue log and source decision — 7 minutes

Notebook Sections 8–9

Add the remaining issues to the example log.

For each, explain the possible research consequence and next action.

Write a short source decision. Then export the documentation bundle.

Raw-file hash: unchanged bytes should retain the same fingerprint.

Your project dataset — 20 minutes

Notebook Section 10

1. State your question and the intended use of the source.
2. Record provenance, definitions, coverage, version and terms.
3. Import an approved CSV or audit restricted-source metadata.
4. Define and interpret at least three relevant checks.
5. State your readiness decision and next action.

Peer check and individual revision — 7 minutes

Work in **five pairs**.

- **3 minutes each:** trace one claim to metadata, examine one check, and identify one unresolved issue.
- **1 minute individually:** record a revision or justified decision to retain the design.

Peer feedback should identify a specific improvement.

Save and submit — 3 minutes

Save your notebook and source documentation.

Confirm that you have:

- written interpretations and an issue log
- a source decision and next action
- a peer-feedback revision
- an AI-use statement and verification note

Follow the posted **Cyber Campus** submission instructions.

Appendix – tools, templates, and follow up

Notebook map

Section	Purpose	Status
1–3	Setup, source note, import	Core practice
4	Data dictionary	Core practice
5–7	Types, validation, record inspection	Core practice
8–9	Issue log, decision, export	Core practice
10	Individual project audit	Required project application
11	Live World Bank API	Optional, off by default
12	Reproducibility and submission	Final check

Software and responsibilities

Task	Tools	What you must still decide
Source discovery	Portal search, literature, AI assistance	Whether the actual dataset fits
Inspection	Excel / Sheets, pandas	Unit, type, missing-code meaning
Reproducible access	Colab / Jupyter, Python urllib	Query, coverage and metadata
Documentation	Notebook, CSV dictionary, source log	Evidence and limitations
Version tracking	Dated files, hashes, GitHub	What can be shared publicly

A minimum source log

- Dataset title, original producer, distributor, table/series ID and URL
- Unit, population/geography, frequency and observation period
- Release/update/version if supplied, plus retrieval date
- Access route, exact query/filters, raw file and hash
- License/permissions link, attribution and sharing conditions
- Known limitations and source decision

Three checks for a student project

Project	Possible checks
Vietnam investment and exports	Sector definition, price basis, country-year coverage
Regional inequality and happiness	Matching geography/year, scale definitions, survey coverage
Housing and policy	Transaction versus asking price, repeated listings, exact dates
AI and learning	Student-task IDs, item nonresponse, measurement validity

Choose rules based on your actual dataset and intended claim.

Optional API example

The notebook requests:

Countries: Korea and Vietnam

Period: 2019–2023

Measure: GDP per capita, constant 2015 US dollars

It saves raw response pages, a retrieval log, and an inspection table.

Set `RUN_LIVE_API=True` only when you want a network request.

Reproducibility and troubleshooting

Undefined variable: run earlier setup cells.

File not found: inspect the working folder and uploaded filename.

Unexpected missing identifier: check the importer's missing-code rules.

API unavailable: complete the core lab; retain the error information.

Run All succeeds with failed data checks: inspect the validation report. The file may still need source review.

Methods and software references

- World Bank Indicators API
- World Bank API call structures
- GDP per capita indicator and source metadata
- World Bank dataset terms
- pandas CSV import documentation
- Google Colab FAQ

Practice table and source cards: fictional materials created for this course.